(Please write your Exam Roll No.) *Exam Roll No. ....*

END TERM EXAMINATION

FIFTH SEMESTER (B.TECH) JANUARY-2024

Paper Code: CIC-307	*Subject: Computer Networks*

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including question no.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Attempt **(Any Five)** **(3x5=15)**

a) What is the purpose of a switch in a network?
b) Explain the difference between half-duplex and full-duplex communication.
c) Explain the purpose of a subnet mask in IP networking.
d) Define the role of a DNS server in network communication.
e) Define the term "fiber optic cable" and explain its advantages in data transmission.
f) How flow and error control works in data link layer?

UNIT-I

Q2 a) What is OSI model. Explain its layers in detail with the help of diagram. Also list the difference between OSI and TCP/IP model? **(10)**

b) An 8 bit byte binary value 10101111 is to be encoded using an even parity Hamming code. What is the binary value after encoding? **(5)**

Q3 a) Explain different types of Guided and Unguided transmission media with the help of diagram. Also list out the advantages and disadvantages of each? **(10)**

b) What is Packet Switching? Explain Virtual circuit and Datagram approach in detail? **(5)**

UNIT-II

Q4 a) What are the various design issues in Data Link Layer? Explain the Stop and Wait ARQ protocol with diagram. **(8)**

b) Compare and contrast HDLC and PPP. Explain the frame format of both with a diagram. **(7)**

Q5 a) Compare and contrast the features of repeaters, hubs, switches, and bridges. Describe the functions of each device and emphasize how each one manages traffic flow and improves network performance. **(8)**

b) Explain channel allocation problem and its solution. Provide an example to illustrate. **(7)**

UNIT-III

Q6 a) What is the difference between classful and classless IP addressing? Explain using an example. **(5)**

b) Explain the Leaky bucket algorithm and illustrate how traffic congestion can be reduced. **(5)**

c) What is the maximum number of subnets in each case? **(5)**

i) Class A; mask 255.255.192.0
ii) Class B; mask 255.255.192.0
iii) Class C; mask 255.255.255.192
iv) Class D; mask 255.255.255.240

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Q7 a) What are distance-vector and link-state routing algorithms. Explain? **(5)**

b) Describe the role of ARP in computer networks. How does ARP help devices communicate within the same network? Provide a basic explanation of the ARP process. **(5)**

c) An organisation is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets. **(5)**

i) Find the subnet mask.
ii) Find the number of addresses in each subnet.
iii) Find the first and last addresses in subnet 1.
iv) Find the first and last addresses in subnet 32.

UNIT-IV

Q8 a) Explain the differences between UDP and TCP in the transport layer. Discuss their main characteristics and provide examples where using UDP might be more advantageous than TCP? **(7.5)**

b) Describe the importance of congestion control in the transport layer. How does congestion control contribute to maintaining network stability and ensuring a reliable data transfer process? **(7.5)**

Q9 a) Explain the client-server model in the context of the application layer. Discuss the roles of clients and servers in this model, highlighting how communication is established between them? **(7.5)**

b) Provide an overview of the Domain Name System (DNS). How does DNS work to translate domain names into IP addresses? Explain the role of DNS servers in facilitating web browsing and internet communication? **(7.5)**

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